

ONEFOUR, AB, Canada

WMO: 711160

Lat: 49.1231N

Lon: 110.4689W

Elev: 3068

StdP: 13.14

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.4	-14.4	-26.4	1.4	-20.4	-19.9	2.1	-14.1	39.9	17.6	36.0	19.8	11.0	310	0.751

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.2	91.7	61.5	88.1	60.7	84.3	59.9	65.1	81.9	63.4	80.6	61.7	78.9	10.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.0	86.8	69.0	57.7	79.8	67.4	55.6	73.9	65.9	31.7	81.5	30.3	80.6	29.0	78.8	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.8	27.8	24.6	-26.4	97.3	5.3	3.1	-30.2	99.5	-33.3	101.4	-36.3	103.1	-40.2	105.4		
			-26.5	68.3	5.3	2.6	-30.3	70.2	-33.4	71.7	-36.4	73.2	-40.2	75.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	16.1	18.7	29.4	41.8	51.9	60.0	68.3	66.7	56.6	43.0	29.1	18.1	
	DBStd	21.51	15.89	15.17	13.99	9.43	8.08	6.52	6.34	6.85	8.84	9.60	13.12	14.60	
	HDD50	4824	1050	876	640	272	74	5	0	0	35	254	630	988	
	HDD65	8793	1514	1296	1103	698	412	177	38	65	276	684	1077	1453	
	CDD50	1818	0	0	2	25	131	304	567	517	233	36	3	0	
	CDD65	311	0	0	0	0	4	26	140	117	24	0	0	0	
	CDH74	4676	0	0	0	11	144	424	1986	1650	445	16	0	0	
	CDH80	1852	0	0	0	1	28	123	858	689	153	0	0	0	
Wind	WSAvg	11.6	12.1	11.5	12.1	12.7	12.4	11.7	11.0	10.7	10.8	11.5	11.1	11.3	
Precipitation	PrecAvg	13.7	1.0	0.6	1.0	1.0	2.0	2.3	1.3	1.4	1.3	0.6	0.6	1.0	
	PrecMax	21.9	2.7	2.9	4.1	3.5	5.6	8.0	2.8	3.3	7.2	1.7	1.7	2.8	
	PrecMin	6.9	0.2	0.0	0.1	0.0	0.2	0.1	0.0	0.1	0.2	0.0	0.0	0.2	
	PrecStd	3.7	0.8	0.6	0.8	0.7	1.3	1.7	0.8	0.8	1.5	0.5	0.5	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	54.2	64.7	74.9	83.4	89.8	96.4	95.6	90.1	76.4	63.6	50.2	
		MCWB	41.7	42.0	47.8	52.4	57.5	61.8	63.1	61.8	59.2	54.8	48.3	39.9	
	2%	DB	45.4	47.0	58.1	68.5	78.1	83.1	92.7	91.0	84.5	68.8	56.3	44.9	
		MCWB	38.3	38.1	44.5	50.2	56.0	59.8	62.5	60.3	57.5	51.4	44.1	37.2	
	5%	DB	41.6	42.2	52.8	63.0	73.1	78.7	88.9	87.6	79.3	63.2	51.0	41.2	
		MCWB	36.1	35.4	41.3	46.9	53.8	58.8	61.9	59.9	56.2	48.8	41.1	35.1	
	10%	DB	37.7	38.0	47.4	58.1	68.2	74.8	84.7	83.6	73.6	58.3	46.1	37.6	
		MCWB	33.5	33.1	38.5	44.4	51.4	57.4	61.2	59.4	54.2	46.4	38.3	32.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.5	42.6	48.2	54.0	59.9	66.3	69.2	65.5	62.6	57.5	49.9	41.5
			MCDB	49.7	53.2	62.9	70.4	77.2	80.1	85.9	82.4	81.9	66.6	60.4	48.0
2%		WB	38.9	38.6	45.0	50.9	57.3	63.0	66.1	63.6	59.5	52.5	45.0	37.9	
		MCDB	44.6	46.0	56.2	66.2	73.4	77.0	82.7	81.7	77.8	66.5	55.2	44.1	
5%		WB	36.2	35.9	42.0	48.3	55.2	60.7	64.3	62.0	57.3	49.6	41.7	35.3	
		MCDB	41.4	41.7	51.7	61.8	70.5	74.4	82.2	80.5	74.8	61.8	50.0	40.8	
10%		WB	33.5	33.2	39.0	45.4	52.9	58.5	62.6	60.5	55.2	46.9	38.6	32.7	
		MCDB	37.8	37.7	46.9	56.5	65.8	71.3	80.0	79.1	72.2	57.0	45.4	37.5	
Mean Daily Temperature Range	5% DB	MDBR	17.9	18.4	19.5	22.2	23.1	22.8	27.2	27.2	25.4	22.6	19.4	17.7	
		MCDBR	22.2	22.9	27.3	31.3	31.2	29.7	33.2	34.2	33.9	30.7	26.0	22.1	
		MCWBR	16.3	15.5	15.4	16.2	14.2	12.3	11.7	12.1	14.2	15.9	16.0	15.6	
	5% WB	MCDBR	22.5	22.5	26.5	29.8	27.9	26.3	28.6	28.7	30.8	28.5	24.8	21.6	
		MCWBR	17.0	16.0	15.7	15.8	13.3	12.0	11.6	11.1	13.7	15.5	15.9	15.7	
Clear-Sky Solar Irradiance	taub	0.264	0.262	0.285	0.334	0.349	0.345	0.346	0.366	0.322	0.290	0.261	0.247		
	taud	2.431	2.499	2.492	2.359	2.358	2.426	2.433	2.352	2.447	2.529	2.514	2.450		
	Ebn at Noon	259	290	298	290	289	290	288	276	280	273	260	251		
	Edh at Noon	20	23	28	36	37	35	34	36	29	23	18	17		
All-Sky Solar Radiation	RadAvg	409	717	1125	1561	1858	2074	2255	1818	1297	812	463	320		
	RadStd	34	60	86	99	141	121	117	113	102	77	32	29		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page